

Ethical Considerations and Data Compliance

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Overview

- Introduction to Data Stewardship and Ethics
- PIPEDA and Privacy Compliance (OPC Canada)
- CIHI Health Data Ethics Framework
- Data Ethics 101
- OECD Principles on AI
- ISO/IEC 27001
- NIST Privacy Framework
- Comparing the Frameworks

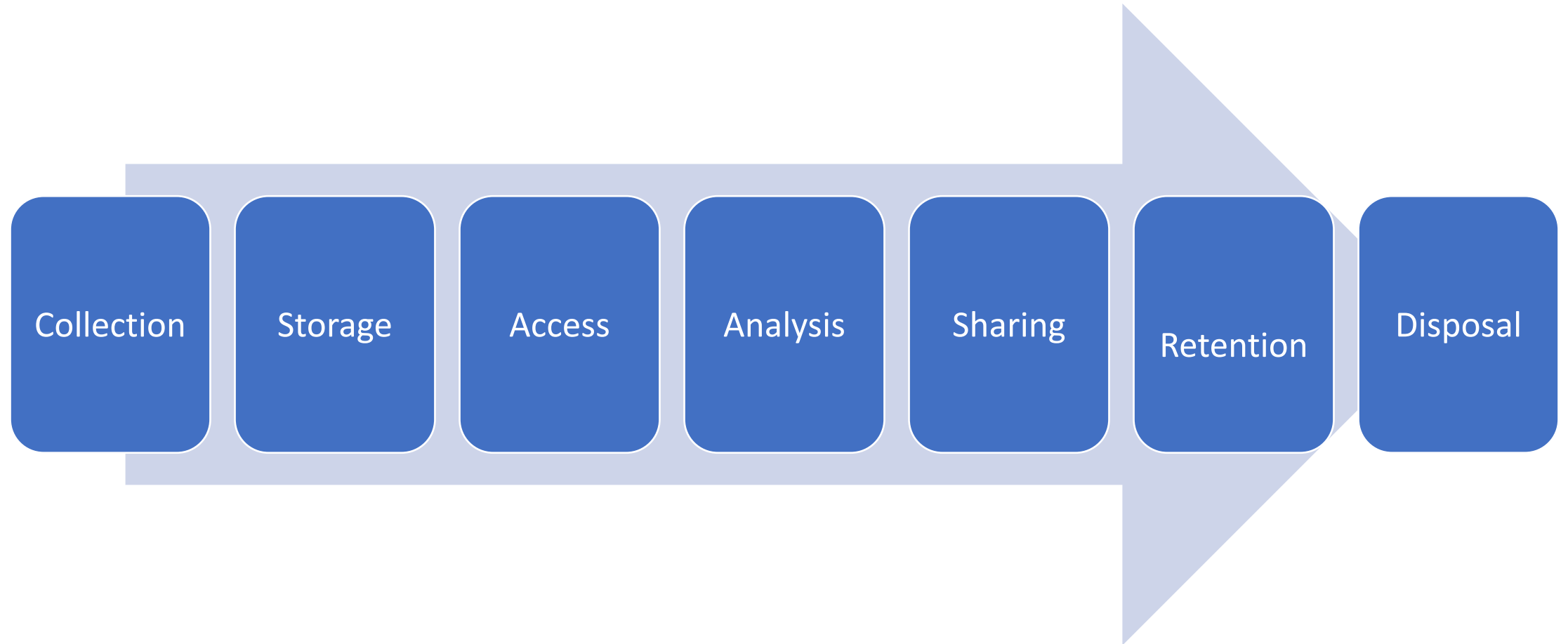
Learning Objectives

- Explain data stewardship principles
- Apply privacy and security frameworks
- Evaluate ethical AI practices
- Compare governance frameworks
- Analyze real-world case studies

Why Data Stewardship Matters

- Supports decision making
- Builds trust
- Reduces risk
- Enables AI governance

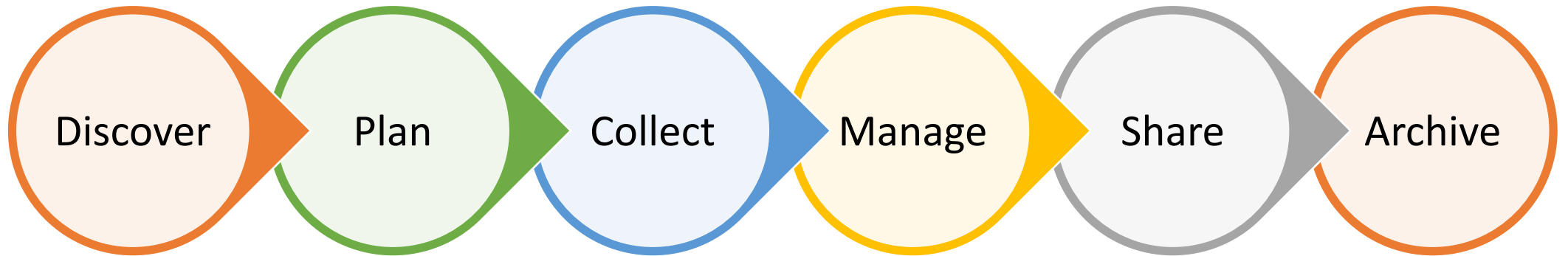
Data Lifecycle Diagram



Government of Canada: Be Good Data Stewards

- Manage data across its lifecycle
- Protect privacy and security
- Maintain data quality and transparency
- Example: Encrypt student records

GoC Stewardship Process



PIPEDA (OPC Canada)

• **PIPEDA** stands for **Personal Information Protection and Electronic Documents Act**. It is Canada's federal privacy law governing how private-sector organizations collect, use, and disclose personal information during commercial activities.

- Accountability
 - Consent
 - Safeguards
 - Openness
- Example: Obtain consent before collecting health survey data

YouTube Video

- [Understanding PIPEDA: Canadian Privacy Law](#)
[Explained – Canadian Privacy](#)

PIPEDA Case Study

Fitness app collects location data without notice.

Discuss consent and transparency.

CIHI Health Data Ethics Framework

• **CIHI** stands for **Canadian Institute for Health Information**. It is an independent, not-for-profit organization that provides standardized health data and information to support improvements in health care, health system performance, and population health across Canada.

- Public benefit
- Protection from harm
- Privacy, confidentiality, and equity

- Example: Use de-identified patient data for research

CIHI Ethical Principles

- Equity
- Transparency and accountability
- Social acceptability
- Example: Check AI models for bias across patient groups

CIHI Health Data Stewardship Toolkit

- Data governance
- Data management
- Public trust and engagement
- Example: Create data access approval workflows

Canada School of Public Service: Data Ethics 101

- Privacy and security
- Fairness and inclusion
- Responsible data use
- Example: Remove demographic bias from datasets

OECD AI Principles

OECD stands for Organization for Economic Co-operation and Development. It is an international organization that works with countries around the world to develop policies that improve economic and social well-being.

- Human-centered values
- Transparency
- Robustness and safety

OECD AI Example

- **Example: Explain how an AI admission system makes decisions**

Suppose a university uses AI to help evaluate applications. The AI might consider factors such as:

- Academic grades
- Prerequisite courses
- English language proficiency
- Portfolio scores
- Program requirements
- Under OECD principles,

The university should be able to explain:

-  What data the AI used
-  What factors influenced the decision
-  Whether a human reviewed the recommendation
-  How students can appeal or question a decision

ISO/IEC 27001

ISO/IEC 27001 is the internationally recognized standard for **Information Security Management Systems (ISMS)**. It provides a framework for organizations to establish, implement, maintain, and continually improve information security practices. [iso.org, en.wikipedia.org]

- ISO = International Organization for Standardization
- IEC = International Electrotechnical Commission
- 27001 = The specific standard for Information Security Management Systems (ISMS)

ISO/IEC 27001

It includes:

- Information Security Management System
- Risk assessment and controls
- Continuous improvement

ISO/IEC 27001

Example:

Multi-factor
authentication
(MFA) for staff

How MFA Supports ISO/IEC 27001?

Suppose a university stores:

- Student records
- Grades
- Research data
- Employee information

A password alone may be stolen through phishing attacks.

MFA requires:

- Something the user knows e.g. Password
- Something the user has e.g. Phone authentication app or Security token
- Something the user is e.g. Fingerprint or Face recognition

NIST Privacy Framework

- The **NIST** Privacy Framework is a voluntary framework developed by the **National Institute of Standards and Technology (NIST)** to help organizations identify, assess, and manage privacy risks while protecting individuals' personal information.
- It takes a risk-management approach and can be used by organizations of any size or sector.

NIST Privacy Framework

Example: Privacy impact assessment before deployment

Function	Purpose
Identify-P	Understand personal data, systems, and privacy risks
Govern-P	Establish privacy policies, roles, and accountability
Control-P	Manage data processing, consent, and privacy choices
Communicate-P	Inform stakeholders how data is used and protected
Protect-P	Safeguard personal information from misuse or unauthorized access

Framework Comparison

- **Stewardship:** GoC, CIHI Toolkit
- **Privacy:** PIPEDA, NIST
- **Ethics:** CIHI, OECD
- **Security:** ISO 27001

Integrated Governance Model

Privacy + Security + Ethics + Stewardship = Trusted Ecosystem

Key Takeaways

- Data must be secure and ethical
- Trust depends on transparency
- AI requires accountability
- Use frameworks together for governance

References

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- National Institute of Standards and Technology. (2020). *Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management*. <https://doi.org/10.6028/NIST.CSWP.01162020>
- [International Organization for Standardization](https://www.iso.org). (2022). *ISO/IEC 27001 Information Security Management Systems*.

Recommended YouTube Videos



[Understanding PIPEDA: Canadian Privacy Law Explained – Canadian Privacy](#)



[AI Ethics: Avoiding Bias and Ensuring Fairness](#)



[Data Governance Explained: A Complete Guide for 2026](#)



Privacy by Design: Principles and Applications – Ann Cavoukian (Recorded Podcast
<https://www.youtube.com/watch?v=Gq5sKZlGxw8>)